

Number of Possible Keys in Playfair Cipher

11. Write a C program for possible keys does the Playfair cipher have? Ignore the fact that some keys might produce identical encryption results. Express your answer as an approximate power of 2.

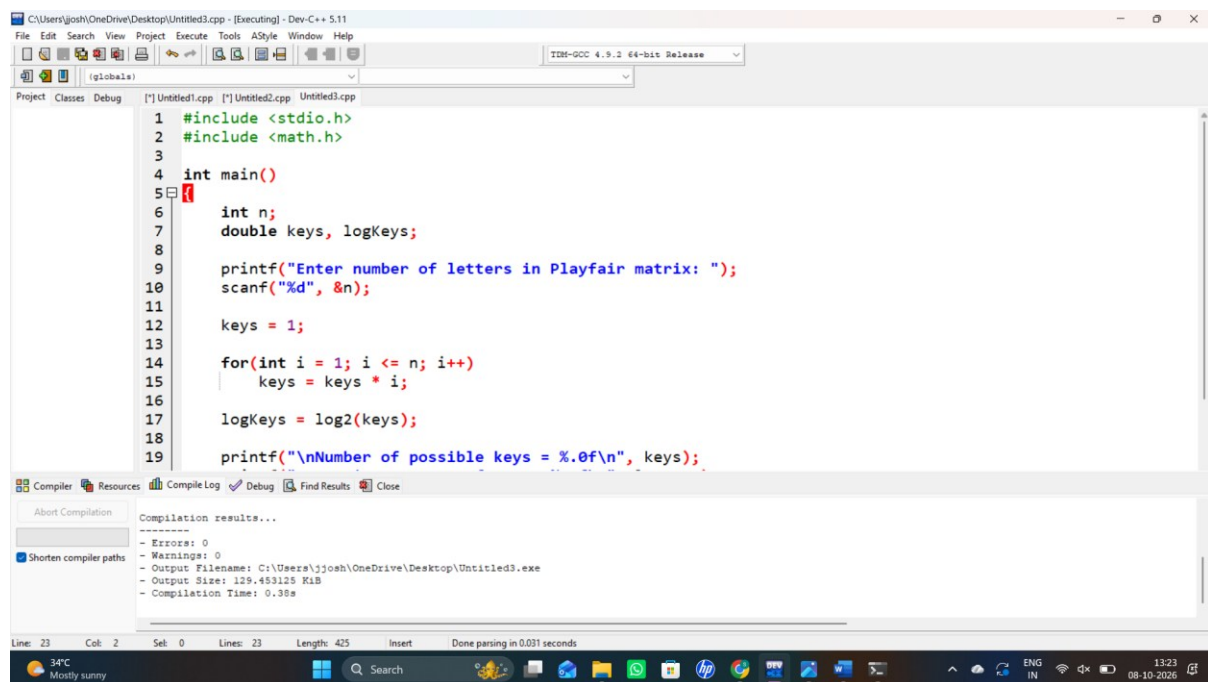
Aim

To determine the approximate number of possible keys in the Playfair cipher and the number of effectively unique keys.

Algorithm

1. A Playfair key is a 5×5 matrix containing 25 letters.
2. The number of possible arrangements is approximately: $\lfloor 25! \rfloor$
3. Express $25!$ as a power of 2.
4. Since some keys produce identical encryption results due to equivalent transformations, calculate the effective number of unique keys.
5. Display both values.

Program

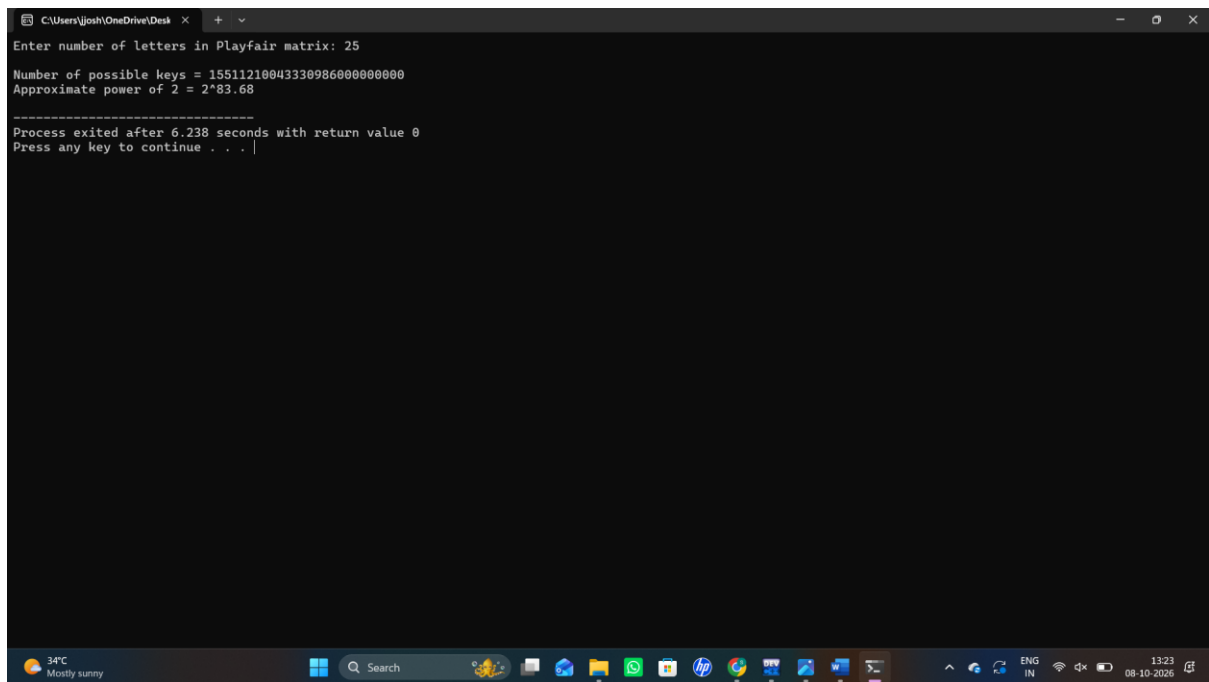


```
1 #include <stdio.h>
2 #include <math.h>
3
4 int main()
5 {
6     int n;
7     double keys, logKeys;
8
9     printf("Enter number of letters in Playfair matrix: ");
10    scanf("%d", &n);
11
12    keys = 1;
13
14    for(int i = 1; i <= n; i++)
15        keys = keys * i;
16
17    logKeys = log2(keys);
18
19    printf("\nNumber of possible keys = %.0f\n", keys);
20 }
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled3.exe
- Output Size: 129.453125 KiB
- Compilation Time: 0.38s

Input/Output



```
C:\Users\josh\OneDrive\Desktop >
Enter number of letters in Playfair matrix: 25
Number of possible keys = 15511210043309060000000000
Approximate power of 2 = 2^83.68
-----
Process exited after 6.238 seconds with return value 0
Press any key to continue . . . |
```

Result

Thus, the number of possible Playfair keys is approximately:

[$25!$ approx $2^{\{83.68\}}$ approx $2^{\{84\}}$]